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BE (6th Semester)

Examination, April-May, 2016

(New Scheme)

Compiler Design

Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) Attempt all questions. Part (*a*) of each question is compulsory. Solve any **two** parts from (*b*), (*c*) and (*d*) of each question.

(*ii*) The figures in the right-hand margin indicate marks.

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Unit - I

1. (*a*) Define Compiler and Interpreter. 2

(*b*) Explain the phases of compiler with each part and suitable diagram for their implementation steps. 7

(*c*) Draw and explain the transition diagram and corresponding code that recognizes :

(*i*) Identifier

(*ii*) Relational operators 7

(*d*) How the compiler can be divided in different parts? Explain the role of single pass and multipass compiler. 7

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Unit - II

2. (*a*) Define Context Free Grammars. 2

(*b*) What is ambiguous grammar ? Explain how to eliminate ambiguous grammar with suitable example. 7

(*c*) What is shift and reduce ? Explain the implementation of shift and reduce with suitable example with steps. 7

(*d*) Define LR Parser and write the method to construct LR Parser table using suitable expression grammar. 7

Unit - III

3. (*a*) Define syntax directed translation. 2

(*b*) Differentiate between a syntax tree and parse tree with example. 7

(*c*) Explain about the syntax directed translation schema for the Boolean expression with examples. 7

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(*d*) Translate the following statement

$A = - B * (C + D)$ to

(*i*) Three Address Statement

(*ii*) Quadraples representation

(*iii*) Direct and Indirect triple representation 7

Unit - IV

4. (*a*) What are activation records ? 2

(*b*) Explain the working of procedures and its activations and run-time. Write with suitable example. 7

(*c*) Give a brief explanation on static allocation and dynamic allocation with respective diagram. 7

(*d*) Discuss the role of symbol table in run-time environment with example. 7

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Unit - V

5. (*a*) Discuss Code Generator. 2

(*b*) What is loop optimization ? Give the suitable techniques and reasons to get benefits from loop optimization. 7

(*c*) Explain in detail issues arises in the design of code generator. 7

(*d*) Write short notes on : 7

(*i*) Global data flow analysis

(*ii*) Error recovery techniques in parsing

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